

RSA Algorithm

Module III

RSA Algorithm

- **RSA** (Rivest-Shamir-Adleman) is an algorithm used to encrypt and decrypt messages.
- This algorithm was described in 1977.
- It is an asymmetric cryptographic algorithm. Asymmetric means that there are two different keys.
- This is also called as public key cryptography, since one of the keys can be given to anyone.

RSA Algorithm

Public key generation

- **Steps:** generate public and private key, encrypt using public key and decrypt using the private key.
- **Rules for generating a public key:**
- Select two prime numbers p and q .
- Let $n = p * q$
- Select a number e such that e is $1 < e < \phi(p) * \phi(q)$ and $\gcd(e, \phi(p) * \phi(q)) = 1$.
- The public key is $\{ e, n \}$.

Example

- Select $p=3$ and $q=11$.
- Let $n = p * q = 3 * 11 = 33$
- Select a number 13 such that 13 is $1 < 13 < 2 * 10$ and $\gcd(13, 20) = 1$.
- The public key is $\{ 13, 33 \}$.

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Private key generation

- **Rules for generating a private key:**
- Find a number d such that,
$$e * d \bmod \phi(p) * \phi(q) = 1$$
- d is a multiplicative Inverse here.
- When you will have multiplicative Inverse for a number in modular arithmetic $e * d \bmod n = 1$, only when e and n are co-prime.
- The private key is $\{d, n\}$.

Example

- Find d ,
- $e * d \bmod \phi(p) * \phi(q) = 1$
- $13 * d \bmod 20 = 1$, $d=17$.
- The private key is $\{17, 33\}$.

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Encryption

- The public key is { 13, 33}. // The public key is { e, n}.
- The private key is { 17, 33}. // The private key is { d, n}.
- Encrypt plain text p=4. (Important $p < n$)
- $\text{cipher} = p^e \bmod n$
- $\text{cipher} = 4^{13} \bmod 33 = 31$

Decryption

- The private key is { 17, 33}. // The private key is { d, n}.
- Decrypt the cipher text 31. (Important $p < n$)
- $\text{plaintext} = C^d \bmod n$
- $\text{plaintext} = 31^{17} \bmod 33 = 4$

- https://en.wikipedia.org/wiki/RSA_Factoring_Challenge

What are the prime factors for this large number (250 digit decimal number)?

- 2140324650240744961264423072839333563
0086147151447550177977549208814180234
4714013664334551909580467961099285187
2470914587687396261921557363047454770
5208051190564931066876915900197594056
9345745223058932597669747168173806936
4894699871578494975937497937

What are the prime factors for this large number (250-digit decimal number)?

- 21403246502407449612644230728393335630086147151447550177977549208
81418023447140136643345519095804679610992851872470914587687396261
92155736304745477052080511905649310668769159001975940569345745223
0589325976697471681738069364894699871578494975937497937

RSA-250 =

6413528947707158027879019017057738908482501474294344720811685963202
4532344630238623598752668347708737661925585694639798853367

*

3337202759497815655622601060535511422794076034476755466678452098702
3841729210037080257448673296881877565718986258036932062711